



Linear Programming

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EgyE 332

1s AY 2026-2027

Outline

1. Motivational Example
2. Simplex Method
3. Examples in Energy Engineering
 - a. Economic Dispatch LP (with and w/o Reserve)
 - b. Economic Dispatch LP with Renewables
 - c. Fuel Scheduling with Inventory

Linear Programming

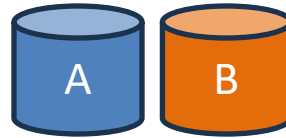
Motivational Example: Fuel Blending

https://github.com/kspilario/EgyE332_opti/blob/main/LP%20Problems/simple_fuel_blending.ipynb

A plant can use two Fuels A and B in which:

x_1 = tons of Fuel A used

x_2 = tons of Fuel B used



Fuel A provides a benefit of PhP 30.00 per ton, while Fuel B provides a benefit of PhP 40.00 per ton.

The plant wants to maximize total benefit, but it is subject to two limits: a **sulfur limit** and **fuel budget limit** (See table on the right).

The maximum allowed sulfur index is 180, while the maximum fuel budget is PhP 2400.00. How many tons of each fuel should be blended within budget?

Property	Fuel A	Fuel B
Benefit per ton	30	40
Cost per ton	20	30
Sulfur index per ton	3	1



$$\begin{aligned} \max \quad & 30x_1 + 40x_2 \\ \text{s. t.} \quad & 3x_1 + x_2 \leq 180 \\ & 20x_1 + 30x_2 \leq 2400 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Mathematical Programs

In general, the method for solving mathematical models is highly dependent on the **type of model**:

TYPE	Objective Function	Constraints	Decision Variables
Linear Program (LP)	Linear	Linear	Continuous
Nonlinear Program (NLP)	At least the objective function or 1 constraint is nonlinear		Continuous
Integer Program (IP)	Linear or nonlinear	Linear or nonlinear	All Integer
Mixed-integer Linear Program (MILP)	Linear	Linear	Some Integer; Others Continuous
Mixed-integer Nonlinear Program (MINLP)	At least the objective function or 1 constraint is nonlinear		Some Integer; Others Continuous



$$\begin{aligned}
 &\max \quad 30x_1 + 40x_2 \\
 &s. t. \quad 3x_1 + x_2 \leq 180 \\
 &\quad \quad 2x_1 + 3x_2 \leq 240 \\
 &\quad \quad x_1, x_2 \geq 0
 \end{aligned}$$

Linear means:

- No products: x_1x_2
- No nonlinear functions: $\sin x_1, \sqrt{x_2}, e^{x_1}, \ln x_1$
- No exponents: x_1^2
- No fractions: $\frac{x_1}{x_2}$

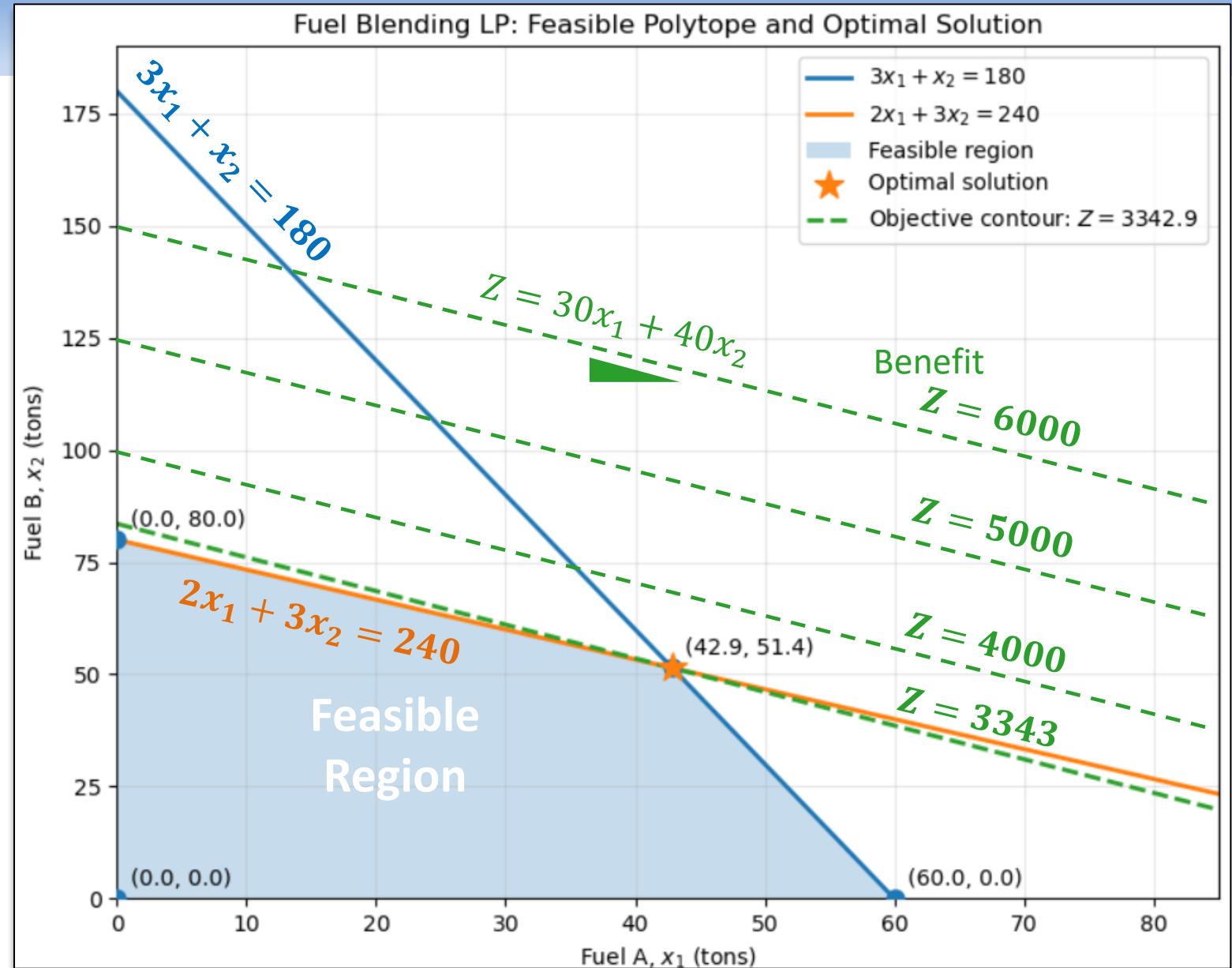
Source: Chap.14; El-Halwagi, M., 2012. *Sustainable Design Through Process Integration*. USA: Elsevier Inc.

*The word "program" is due to George Dantzig (1974).

Simplex Method

Let's visualize the feasible region first:

$$\begin{aligned} \max \quad & Z = 30x_1 + 40x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq 180 \\ & 2x_1 + 3x_2 \leq 240 \\ & x_1, x_2 \geq 0 \end{aligned}$$



Simplex Method

Let's visualize the feasible region first:

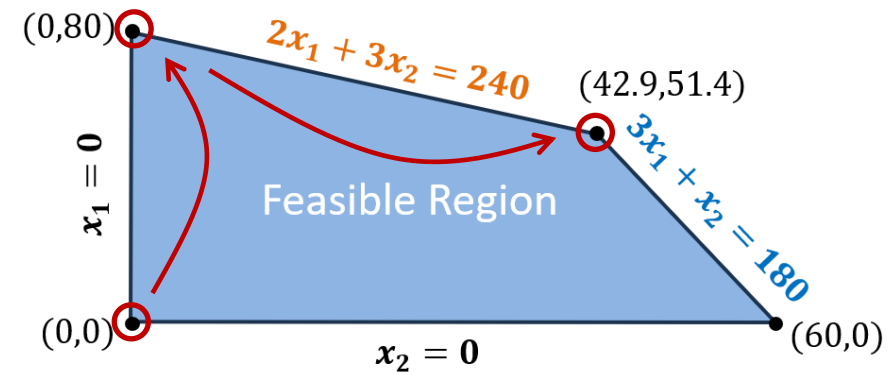
$$\begin{aligned} \max \quad & Z = 30x_1 + 40x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq 180 \\ & 2x_1 + 3x_2 \leq 240 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Introduce slack variables, s_i :

$$3x_1 + x_2 + s_1 = 180$$

$$2x_1 + 3x_2 + s_2 = 240$$

$$x_1, x_2, s_1, s_2 \geq 0$$



Simplex Tableau:

STEP 1	x_1	x_2	s_1	s_2	RHS
s_1	3	1	1	0	180
s_2	2	3	0	1	240
z	-30	-40	0	0	0

STEP 2	x_1	x_2	s_1	s_2	RHS
s_1	7/3	0	1	-1/3	100
s_2	2/3	1	0	1/3	80
z	-10/3	0	0	40/3	3200

STEP 3	x_1	x_2	s_1	s_2	RHS
s_1	1	0	3/7	-1/7	42.9
s_2	0	1	-2/7	3/7	51.4
z	0	0	10/7	90/7	3343

Solution so far:

$$(x_1, x_2, s_1, s_2)$$

$$(0, 0, 180, 240)$$

$$\begin{aligned} R_2 &\leftarrow R_2/3 \\ R_1 &\leftarrow R_1 - R_2 \\ R_3 &\leftarrow R_3 + 40R_2 \end{aligned}$$

$$(0, 80, 100, 0)$$

$$\begin{aligned} R_1 &\leftarrow 3R_1/7 \\ R_2 &\leftarrow R_2 - 2R_1/3 \\ R_3 &\leftarrow R_3 + 10R_1/3 \end{aligned}$$

$$(42.9, 51.4, 0, 0)$$

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Economic Dispatch: Simple

A very simple economic dispatch problem is shown below:

$$\min_{P_g} \sum_g c_g P_g$$

$$\text{s.t. } \sum_g P_g = D$$

$$P_g^{\min} \leq P_g \leq P_g^{\max}$$

Minimize total cost of generating power for 1 hr

Such that the demand is exactly met, and

Each generator g outputs power within bounds.

Sample Data: $D = 300$ MW

g	Generator	c_g	$P_g^{\min}$	$P_g^{\max}$
1	Coal	20	50	200
2	Gas	25	20	150
3	Diesel	40	10	100
		\$/MWh	MW	MW

Notation: P_g = power output of generator g (Decision variables)

c_g = cost per MWh of power output of generator g

$P_g^{\min}, P_g^{\max}$ = min and max power output for generator g

D = total demand

* Assume a single period of 1 hr.

Economic Dispatch: Simple with Reserve

A simple economic dispatch problem with reserve is shown below:

$$\min_{P_g, R_g} \sum_g c_g P_g + \sum_g c_g^R R_g$$

$$s. t. \sum_g P_g = D$$

$$P_g^{min} \leq P_g \leq P_g^{max}$$

$$P_g + R_g \leq P_g^{max}$$

$$\sum_g R_g \geq R^{req}$$

Minimize total cost of generating power for 1 hr

Such that the demand is exactly met, and

Each generator g outputs power within bounds, and

Power output + reserved power are below max, and

Total reserved power is above a required amount.

* Assume a single period of 1 hr.

Sample Data: $D = 300$ MW, $R^{req} = 120$ MW

g	Generator	c_g	c_g^R	P_g^{min}	P_g^{max}
1	Coal	20	8	50	200
2	Gas	25	4	20	150
3	Diesel	40	3	10	100
		\$/MWh	\$/h	MW	MW

Additional Notation:

R_g = reserved power of generator g (Decision variables)

c_g^R = cost of reserving power for generator g per hr

R^{req} = total reserved power required

Fuel Scheduling with Inventory

A simple *multi-period* fuel scheduling problem is shown below:

$$\min_{P_{g,t}, B_{g,t}, M_{g,t}} \sum_t \sum_g c_g B_{g,t}$$

$$s. t. \sum_g P_{g,t} = D_t$$

$$P_g^{min} \leq P_{g,t} \leq P_g^{max}$$

$$M_{g,t} = M_{g,t-1} + B_{g,t} - \frac{h_g P_{g,t} \Delta t}{H_g}$$

$$0 \leq M_{g,t} \leq M_g^{max}$$

Minimize total cost of purchased fuel

Hourly demand balance

Generator capacity limit

Fuel inventory balance

Fuel storage capacity

Sample Data:

g	Generator	h_g	H_g	P_g^{min}	P_g^{max}
1	Coal	10	25	50	100
2	Gas	7	50	20	80
3	Diesel	9	43	10	50

GJ/MWh GJ/ton MW MW

Additional Notation:

B_g = fuel bought for gen g at time t
 M_g = stored fuel for generator g [ton]

(Decision variables)

h_g = heat rate for generator g per [GJ/MWh]

H_g = heating value of fuel for generator g [GJ/ton]

* Assume each Δt is 1 hr and a total period of 24 hrs.

Economic Dispatch: Renewables + Storage

A *multi-period* economic dispatch problem with Solar, Wind, and Battery Storage is given:

$$\min_{P_{g,t}, P_{s,t}, P_{w,t}, P_{ch,t}, P_{dis,t}} \sum_{t=1}^{24} \left[\sum_g c_g P_{g,t} + c_B (P_{ch,t} + P_{dis,t}) \right]$$

$$s.t. \quad P_{s,t} + P_{w,t} + P_{dis,t} + \sum_g P_{g,t} = D_t + P_{ch,t}$$

$$0 \leq P_{s,t} \leq S_t^{avail}$$

$$0 \leq P_{w,t} \leq W_t^{avail}$$

$$P_g^{min} \leq P_{g,t} \leq P_g^{max}$$

$$0 \leq P_{ch,t} \leq P_{ch}^{max}$$

$$0 \leq P_{dis,t} \leq P_{dis}^{max}$$

$$E_t = E_{t-1} + \eta_{ch} P_{ch,t} - \frac{P_{dis,t} \Delta t}{\eta_{dis}}$$

$$0 \leq E_t \leq E^{max}$$

$$E_{24} = E_0$$

Minimize total cost of generating power for 24 hrs,

Such that the demand and stored energy are exactly met by the generators, RE, and the power discharged by battery, and

Solar and wind power utilization is within the available,

Each generator g outputs power within bounds, and

Battery charging and discharging power are within limits, and

Battery storage increase / decrease with some efficiency, and
Battery storage does not exceed a maximum, and
Stored energy at $t = 0$ must match that after 24 hrs.

* Assume each Δt is 1 hr and a total period of 24 hrs.

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